കണ്ണൂർ ജില്ലാ പഞ്ചായത്ത്

പൊതുവിദ്യാഭ്യാസ
വകുപ്പ്- കേരളം

ഡയറ്റ് കണ്ണൂർ

समग्र शिक्षा के रल
Samagra Shiksha Kerala
സമഗ്ര ശിക്ഷാ കേരളം

2



എസ് എസ് എൽ സി
വിദ്യാർത്ഥികൾക്കുള്ള
പഠനപിന്തുണാ സഹായി 2

Mathematics

PART 2

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കണ്ണൂർ ജില്ലാപഞ്ചായത്ത് - പൊതുവിദ്യാഭ്യാസവകുപ്പ്
ഡയറ്റ് കണ്ണൂർ-സമഗ്രശിക്ഷ കേരളം കണ്ണൂർ

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തയ്യാറാക്കിയത്

സ്മൈൽ റിസോഴ്സ് ഗ്രൂപ്പ് അംഗങ്ങൾ

ഏകോപനം

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(Eng. Med.)

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8 Arithmetic Sequences

9 Circles and Angles

10 Arithmetic Sequences and Algebra

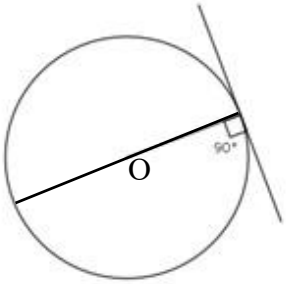
11 Mathematics of Chance

12 Second Degree Equations

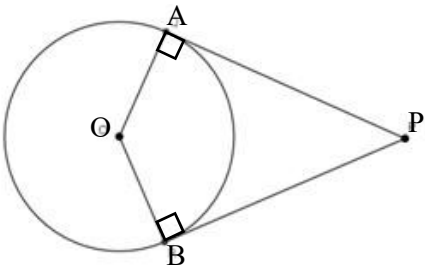
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8. TANGENTS

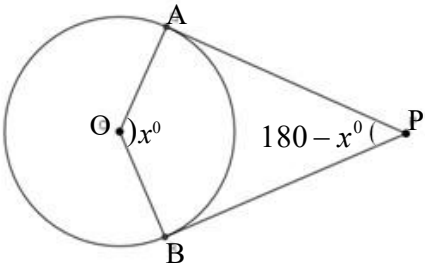
The tangent at a point to a circle is perpendicular to the diameter through that point



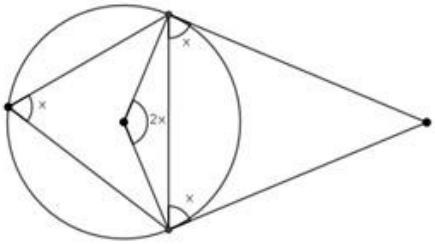
The angle at the centre of a circle between two radii, and the angle between the tangents at the ends of radii add up to 180°



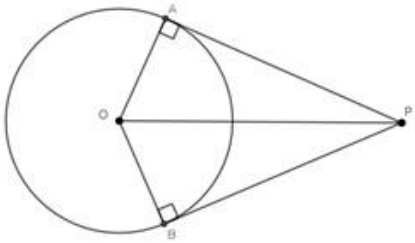
The angle between the tangents at the ends of a chord in a circle is the central angle of the chord subtracted from 180°



The angles which the tangents to a circle at the ends of a chord make on one side of the chord are both equal to the angle in the alternate segment.



The two tangents from a point outside a circle have the same length



In a quadrilateral formed by the tangents at four points on a circle, the sum of the lengths of opposite sides is equal

In any triangle, the bisectors of all three angles intersect at a single point

The radius of the incircle of a triangle is equal to its area divided by half the perimeter

$$r = \frac{A}{s}$$

The area of a triangle with lengths of sides a, b, c is

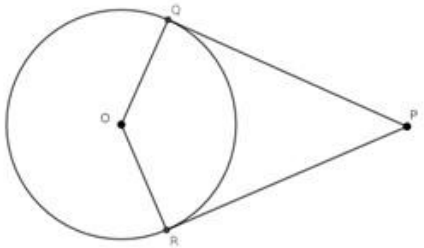
$$\sqrt{\qquad\qquad - c)}$$

The radius of the incircle is this divided by s

$$r = \frac{\sqrt{\qquad\qquad - c)}}{s}$$

More Questions:

1) In the figure Q,R are points on the circle. Tangents passes through these points are intersect at P.



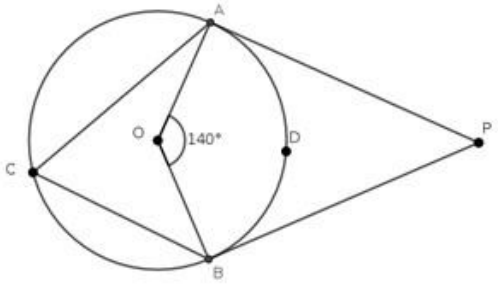
   (*i*) What is $\angle OQP$ and $\angle ORP$?

   (*ii*) Find $\angle OQP$ + $\angle ORP$?

   (*iii*) Calculate $\angle QOR$ + $\angle QPR$.

   (*iv*) If $\angle QOR = 120^o$, find $\angle QPR$.

2) In the figure O is the centre of the circle .A,B ,C are points on the circle . Tangents passes through the points A and B are intersect at P.



   (*i*) What is $\angle APB$?

   (*ii*) What is $\angle ACB$?

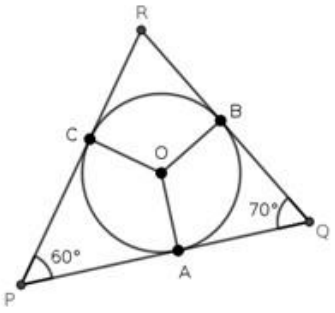
   (*iii*) What is $\angle ADB$?

   (*iv*) If $\angle APB = 60^o$,
   Find $\angle AOB$, $\angle ACB$ and $\angle ADB$.

   (*v*) If $\angle ACB = 50^o$,
   Find $\angle AOB$, $\angle APB$ and $\angle ADB$.

3) In the figure, O is the centre of the circle. Tangents passes through the points A,B,C are intersects at P,Q,R.

$\angle P = 60^o$, $\angle Q = 70^o$



(*i*) Find $\angle R$.

(*ii*) What is $\angle AOC$?

(*iii*) What is $\angle AOB$?

(*iv*) What is $\angle BOC$?

4) Draw a circle of radius 2 centimetres. Mark a point 6 centimetres away from the centre as P. Draw tangents from the point P .

5) Draw a circle of radius 2.5 centimetres and draw tangents to it from a point 7centimetres from the centre.

6) Draw a circle of radius 2 centimetres. Draw an equilateral triangle with its sides touches the circle

7) Draw a circle of radius 2.5 centimetres. Draw a triangle of angles 50°, 60°,70°, with its sides touching the circle.

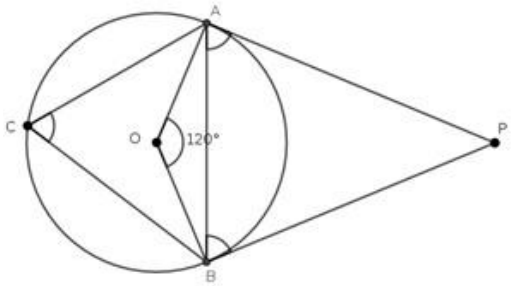
8) Draw a circle of radius 3 centimetres. Draw an isosceles triangle of one angles 40° with its sides touching the circle.

9) Draw a triangle of in radius 2 centimetres and two of its angles are 55°, 75°.

10) Draw the triangle with sides 5 centimetres, 6 centimetres, 7 centimetres and draw its incircle.

11) Draw the triangle with sides 5 centimetres and two of its angles are 65° and 55° and draw its incircle.

12) Iin the figure O is the centre of the circle and PA and PB are tangents to the circle from the point P. $\angle AOB = 120^o$.



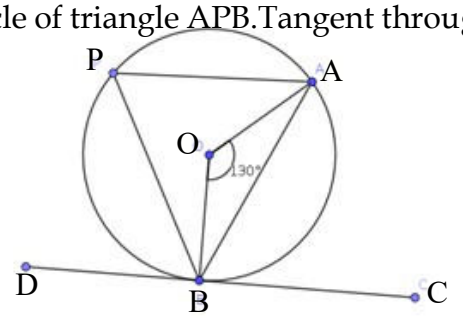
(*i*) Find $\angle ABP$ and $\angle BAP$.

(*ii*) What is $\angle APB$?

(*iii*) What is $\angle ACB$?

13) In the figure O is the centre of circumcircle of triangle APB.Tangent through the point B is DC.

$\angle AOB = 130^o$.



(*i*) What is $\angle OBC$?

(*ii*) What is $\angle APB$?

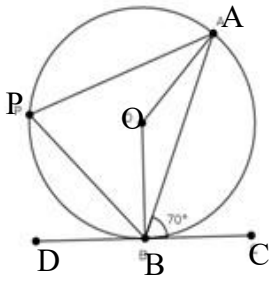
(*iii*) What is $\angle ABC$?

14) In the figure O is the centre of circumcircle of triangle APB.Tangent through the point B is DC.

$\angle ABC = 70^o$.

(*i*) What is $\angle APB$?

(*ii*) What is $\angle AOB$?

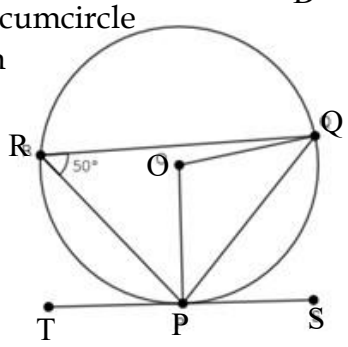


15) In the figure O is the centre of circumcircle of triangle PQR. Tangent through the point P is TS.

$\angle PRQ = 50^o$.

(*i*) What is $\angle POQ$?

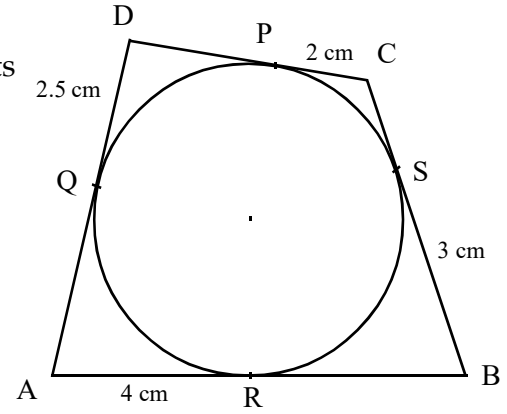
(*ii*) What is $\angle QPS$?



16) Quadrilateral ABCD is formed by the tangents to the circle with centre O.
AR= 4 Centimetres, BS= 3 Centimetres,
CP= 2 Centimetres, QD= 2.5 Centimetres.

(*i*) Find the length of the tangents AQ, BR, CS and DP.

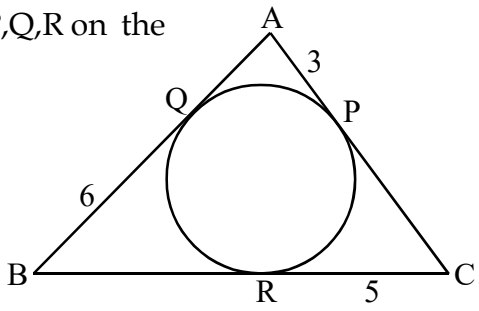
(*ii*) Find the lengths of the sides of the quadrilateral.



17) In the figure, tangents through the points P,Q,R on the circle are the sides of the triangle ABC.

(*i*) Find the perimetre of the triangle ?

(*ii*) Find its circumradius.



18) In the figure A,B,C,D are points on the circle. Tangets through the point C is PQ.

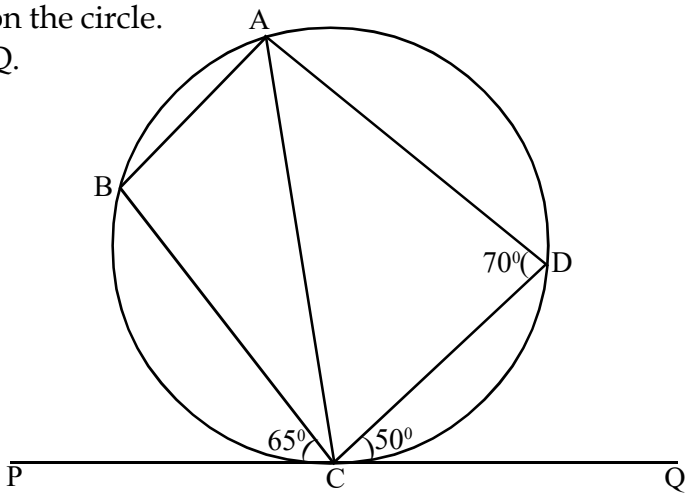
$\angle DCQ = 50^0$, $\angle BPC = 65^0$ and

$\angle D = 70^0$.

(*i*) $\angle CAD =$ _____

(*ii*) $\angle BAC =$ _____

(*iii*) $\angle BCD =$ _____



Polynomials And Equations

For any four numbers x, y, u, v
$(x + y)(u + v) = xu + xv + yu + yv$

Find 37×35

$37 \times 35 = (30 + 7) \times (30 + 5)$

$= (30 \times 30) + (30 \times 5) + (7 \times 30) + (7 \times 5)$

$= 900+150+210+35= 1295$

Otherwise

$37 \times 35 = (30 + 7) \times (30 + 5) = 30^2 + (7+5) \text{ x } 30 + 7x5$

$= 900 + 12x30 + 35$

$= 900 +360 +35 = 1295$

Find 42 x 48

$42 \text{ x } 48 = 40^2 + (2+8)\ 40 + 2x8 = 1600 + 10x40 + 16 = 1600+400+16 = 2016$

We can write the multiplications done in all these examples above as a general principle:

For any number x, $(x + 3)(x + 7) = x^2 + (3 + 7)x + 21= x^2+ 10x + 21$

Eg: Do the multiplication 78 x 72. Write the multiplications done as a general principle

$78x72 = 70^2 + (8+2)\ 70 + 8x2 = 4900 +10x70 + 16 = 4900 +700+16 = 5616$

General principle

$(x+8)\ (x+2) = x^2 +(8+2)x +8x2 = x^2 +10x +16$

Do the multiplication below. Write the multiplications done in all these examples as a general principle

1. 25 x 29 2. 35 x 38 3. 43 x 49 4. 51 x 56

Polynomial multiplication

we have, $(x + 1)(x + 2) = x^2 + 3x + 2$

$(x + 2)(x + 3) = x^2 + 5x + 6$

$(x + 5)(x + 3) = x^2 + 8x + 15$

We can consider all these algebraic expressions as polynomials.

So, the product of the first degree polynomials $x + 1$ and $x + 2$ is the second degree polynomial $x^2 + 3x + 2$.

For any number x

$(x + a)(x + b) = x^2 + (a + b)x + ab$

also we have

$(x + y)(u - v) = xu - xv + yu - yv$

$(x - y)(u - v) = xu - xv - yu + yv$

Using these, we can write the products of some other kinds of first degree polynomials:

- $(x + a)(x - b) = x^2 + (a - b)x - ab$
- $(x - a)(x - b) = x^2 - (a + b)x + ab$

$(x + 3)(x - 2) = x^2 + (3 - 2)x - 3x2 = x^2 + x - 6$

$(x - 5)(x - 4) = x^2 - (5+4)x + 5x4 = x^2 - 9x + 20$

Find the following products

(i) $(x + 1)(x + 4)$

(ii) $(x + 3)(x - 6)$

(iii) $(x - 1)(x + 4)$

(iv) $(x - 3)(x - 4)$

Polynomial factors

Consider a second degree polynomial $x^2 – 9x + 20$

write this second degree polynomials as the product of two first degree polynomials

Let the first degree polynomials be (x+a) and (x+b)

we have $(x + a)(x + b) = x^2 + (a + b)x + ab$

so $x^2 – 9x + 20 = (x + a)(x + b)$

$x^2 – 9x + 20= x^2 + (a + b)x + ab$

we get a+b = -9 and ab = 20

that is product of two numbers is 20 and their sum is -9

20= 1x 20, 20 =-1 x -20 , 20= 2 x 10, 20 = -2 x -10, 20 = 4 x 5, 20 =-5 x -4

from these the sum will be -9 only if -5 and -4 are taken

so first degree polynomial factors will be (x- 5) and (x-4)

$x^2 – 9x + 20 = (x- 5)(x-4)$

Write $x^2 + 34x + 280$ as the product of two first degree polynomials

$x^2 + 34x + 280 = x^2 + (a + b)x + ab$

we get a+b = 34 and ab = 280

we have to factorize 280 as the product of two numbers in different ways, and check which pair gives the sum 34; but this is not very easy. So, let's see if there is a different method

we have $(a − b)^2 = (a + b)^2 − 4ab$

$(a − b)^2 = 34^2 − 4 \times 280 = 1156 − 1120 = 36$

$a-b= \sqrt{36} = 6$

$a + b = 34$

$a − b= 6$

$a= ½(34+6) = 20$, $b= ½(34-6) = 14$

$x^2 + 34x + 280 = (x+20)(x+14)$

Write the polynomials below as the product of two first degree factors

(i) $x^2 + 25x + 150$ (ii) $x^2 + 4x − 192$ (iii) $x^2 − x − 210$

To get $x^2 - 7x + 10 = 0$, what number should we take as x?

Now how do we write the steps leading to the solution?

(i) $x^2 - 7x + 10 = 0$

(ii) $x^2 - 7x + 10 = (x - 5)(x - 2)$

(iii) $(x - 5)(x - 2) = 0$

(iv) $x - 5 = 0$ or $x - 2 = 0$

(v) $x = 5$ or $x = 2$

One side of a rectangle is 4 metres longer than the other side and its area is 221 square metres. What are the lengths of the sides?

Let sides be x and x+4

Area = x(x+4)

so, x(x+4) =221

$x^2 + 4x = 221$

$x^2 + 4x - 221 = 0$

$a + b = 4 \quad ab = -221$

$(a - b)^2 = (a + b)^2 - 4ab = 4^2 - 4 \times -221 = 16+884 = 900$

$a - b = \sqrt{900} = 30$

$a + b = 4$

$a - b = 30$

$a = \frac{1}{2}(4+30) = 17$

$b = \frac{1}{2}(4-30) = -13$

so $x^2 + 4x - 221 = (x+17)(x-13)$

$x^2 + 4x - 221 = 0$

$(x+17)(x-13) = 0$

$x + 17 = 0$ and $x - 13 = 0$

$x = -17$ and $x = 13$

so sides are $x = 13$ and $x+4 = 13+4 = 17$

(1) One side of a rectangle is 3 metres longer than the other and its area is 154 square metres. What are the lengths of its sides?

(2) The sum of the consecutive natural numbers from 1up to which number is 325?

(3) At what points does the graph of the polynomial $x^2 - 4x + 4$ cross the x-axis?

General solution

How to find the solution of the second degree equation of the form $ax^2 + bx + c = 0$ we have a general method to find the solution

$$x = \frac{-b - \sqrt{b^2 - 4ac}}{2a} \text{ or } x = \frac{-b + \sqrt{b^2 - 4ac}}{2a}$$

At what points does the graph of the polynomial $2x^2 + 13x - 15 = 0$ cross the x-axis?

a = 2, b = 13,c = -15

use the equation

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$b^2 - 4ac = 13^2 - 4x2x-15 = 169 + 120 = 289$

$\sqrt{b^2 - 4ac} = \sqrt{289} = 17$

so x= (-13+17)/4 = 1 and (-13-17)/2 = -30/4 = -15/2

x = 1, and -15/2

Some Questions

1. The product of two consecutive positive integers is 240. find the numbers

2. The product of two consecutive even numbers is 288. Find the numbers.

3. Consider the arithmetic sequence 5, 9, 13, · · · What is the position of the term in the sequence whose square is 2025?

4. The smallest side of a right angled triangle is 4 less than its hypotenuse. Third side is 2 more than the smallest side.

a) If the smallest side is x what are the other two sides.

b) Find the length of sides of the triangle.

5. Length of a rectangle is 3 more than its breadth. Area of the rectangle is 418 sq.cm Find the length and breadth of the rectangle .

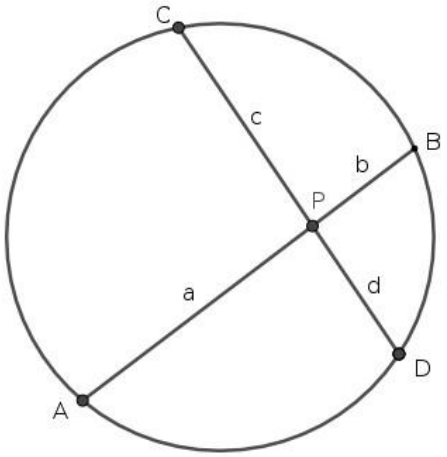
6.The perimeter of a rectangle is 82cm, area 400 sq.cm. Find the sides of the rectangle.

7. The sum of a number and its reciprocal is 25/17. find the number

8. The area of a right triangle is 375 sq.cm. Length of one of the perpendicular sides is 10cm more than the other. Find sides.

9. How many terms of the sequence of natural numbers starting from 1 added to get 465?

CIRCLES AND LINES

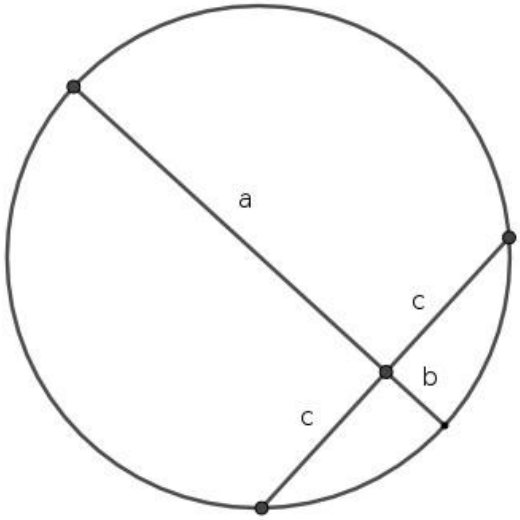


When two chords of a circle intersect within the circle, the product of the parts of one chord is equal to the product of the parts of the other

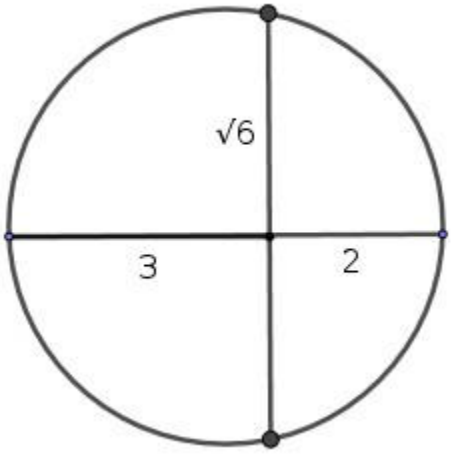
AP × PB = CP × PD

The product of the parts into which a diameter of a circle is cut by a perpendicular chord, is the square of half the chord

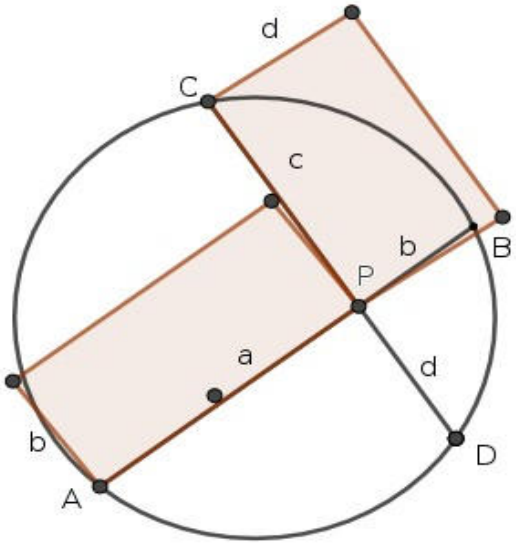
$ab = c^2$



How to draw a line of length √6 using this concept.
We have 3x2 = 6
so draw a circle of diameter 5 cm and divide it in to 3 cm and 2 cm. Draw a perpendicular at the length of 3 cm, it becomes a chord and the
perpendicular is half the chord, and its length is √6.

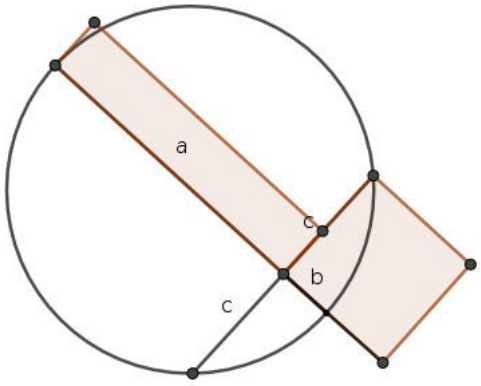


When two chords intersect within a circle, the rectangles with sides as the parts of each chord have equal areas



When a diameter is

perpendicular to a chord, the rectangle with sides as the parts of diameter and the square with side as part of a chord have equal areas.



1. Draw a rectangle of width 6 centimetres and height 4 centimetres.

   (i) Draw a rectangle of the same area with width 7 centimetres.

   (ii) Draw a square of the same area.

2. Draw a square of area 12 square centimetres.

3. Draw a rectangle of sides 4 centimetres and 3 centimetres. Draw a square of the same area.

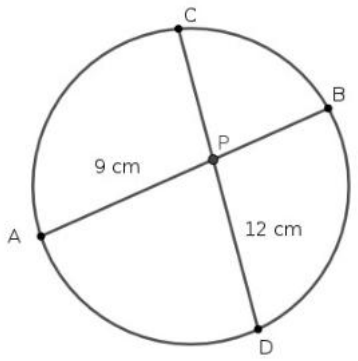
4. Draw a rectangle of area 24 sq.cm. Draw square of area equal to the area of rectangle.

5. The chords AB and CD intersect at P .

AB = 17 cm, PA = 9 cm, PD = 12 cm

a. What is the length of PB

b. Find the length of PC

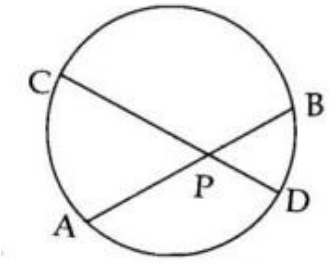


6. Chords AB and CD are intersecting at P.

AB =10 cm, PB = 4 cm and PD =3 cm

a. What is the length of PA?

b. Find the length of PC

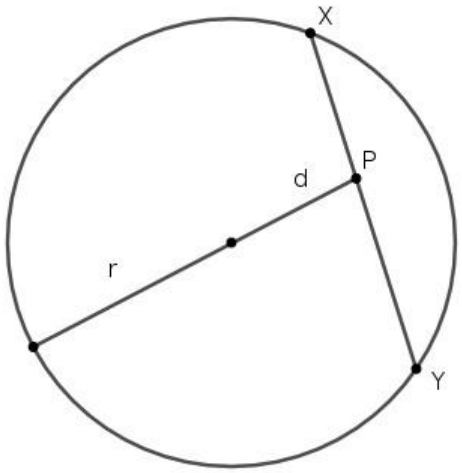


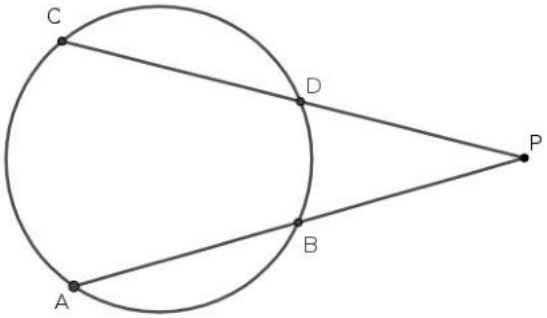
7.Draw a rectangle of area 18 sq.cm. Draw a square of the same area.

8. Draw a square of side $2\sqrt{3}$.

For any chord XY of a circle of radius r, passing through a point P within the circle at a distance d from the centre,

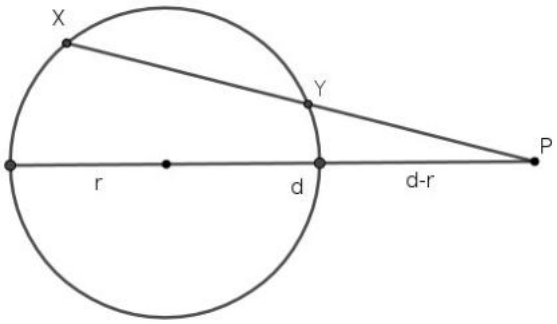
$PX \times PY = r^2 - d^2$





For all lines through a point outside a circle intersecting the circle at two points, the product of the distances from the point to the points of intersection with the circle is the same number

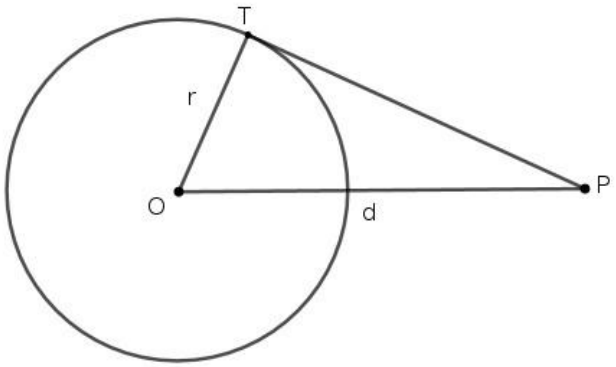
$PA \times PB = PC \times PD$



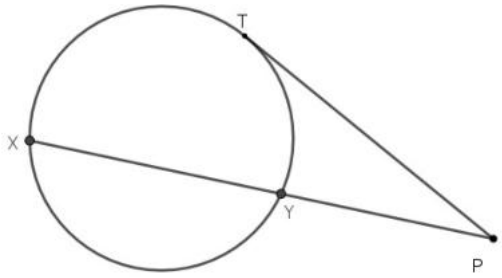
If a line from a point P outside a circle of radius r at a distance d from the centre cuts the circle at X and Y, then

$PX \times PY = d^2 - r^2$

Whatever line we draw from a point outside a circle, intersecting the circle at two points, the product of the distances from the point to the points of intersection is the square of the length of the tangent from the point.



$$d^2 - r^2 = PT^2$$

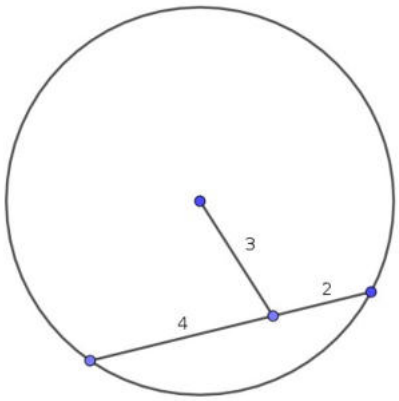


$$PX \times PY = PT^2$$

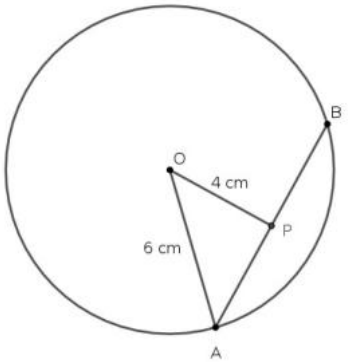
If X and Y are the points of intersection of a circle of radius r with a line through a point P inside or outside the circle, at a distance d from the centre, then

$$PX \times PY = |r^2 - d^2|$$

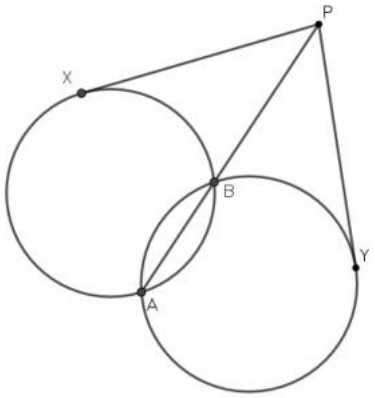
1. In the picture, a line from the centre of a circle cuts a chord into two parts. What is the radius of the circle?



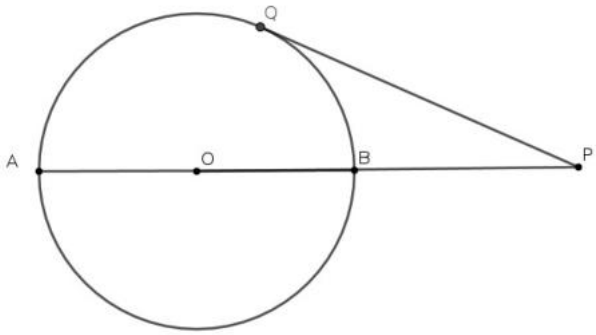
1. In the figure O is the centre of the circle of radius 6 cm. OP= 4 cm, PB= 5cm. Find the length of the chord.



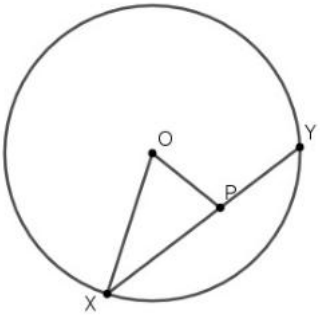
2. In the figure two circles are intersecting at A and B. From a point P from the line joining A and B, two tangents PX and PY are drawn to the circles. PA= 9 cm , PB = 4 cm Find length of the tangents.



3. In the figure AB is the diameter of the circle and PA = 16cm PQ= 8cm. Find PB.



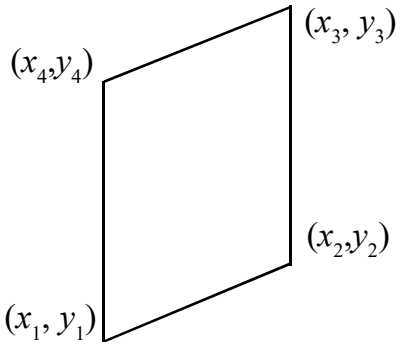
4. In the figure O is the centre of the circle PX= 5 cm, OX=7 cm, OP= 3 cm. Find length of PY



11. GEOMETRY AND ALGEBRA

Parallellograms:

If (x_1, y_1), (x_2,y_2), (x_3, y_3) and (x_4,y_4) are the coordinates of four vertices of a parallellogram.

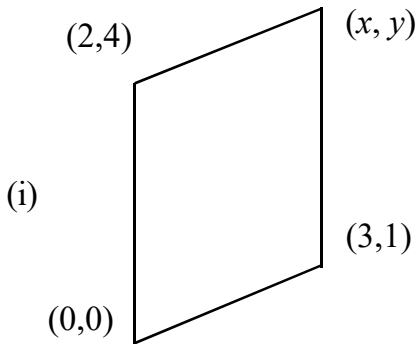


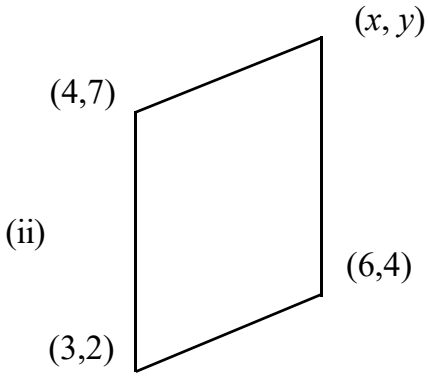
$$x_1 + x_3 = x_2 + x_4$$

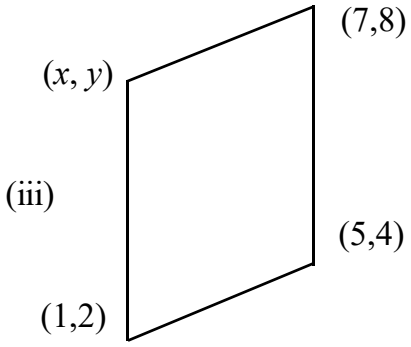
$$y_1 + y_3 = y_2 + y_4$$

The sum of the x- coordinates and the sum of the y- cordinates of opposite vertices of aparallellogram are equal.

♦ In the figure, write (x, y) in the parallellogram.

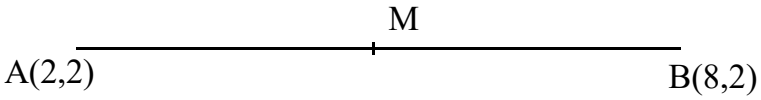






Coordinates of the Midpoints:

The midpoints of the line joining (x_1, y_1) and (x_2, y_2) is $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$



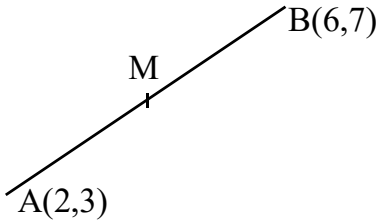
In the figure, M is the midpoints of AB.

The coordinates of M is $\left(\frac{2+8}{2}, 2\right) = (5,2)$

- Write the coordinates of the midpoints of the line joining each of the following pair of points.
  (*i*) (3,1), (11,1)
  (*ii*) (2,3), (8,3)
  (*iii*) $(-1,2)$, (7,2)
  (*iv*) (1,2), (1,6)
  (*v*) $(-3,-5)$, $(-3,1)$

In the figure, M is the midpoints of AB.

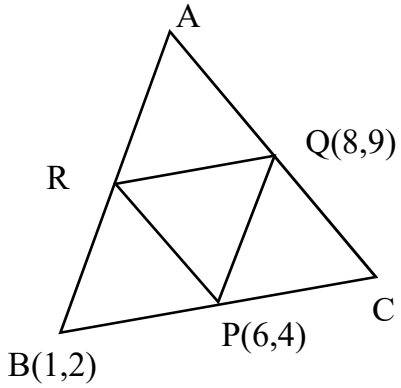
The coordinates of M is $\left(\frac{2+6}{2}, \frac{3+7}{2}\right) = (4,5)$



- Write the coordinates of the midpoints of the line joining each of the following pair of points.
  (*i*) (2,4), (8,10)
  (*ii*) (1,3), (9,7)
  (*iii*) $(-2,1)$, (6,7)
  (*iv*) $(3,-4)$, $(-7,-6)$
  (*v*) $(5,-3)$, $(-1,-5)$

- A circle is drawn with the line joining (6,3) and (4,5) as diameter. What are the coordinates of the centre of the circle.

- In the picture, P,Q and R are the midpoints of the triangle ABC.
  Write coordinates of A,C and R.

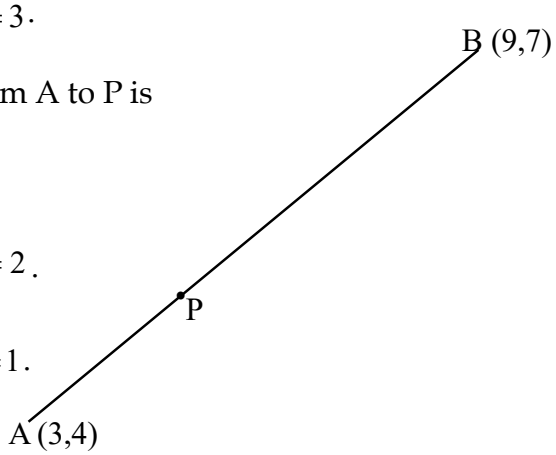


Ratio:

Using the coordinates of two points, we can find the coordinates of the points that devides the line joining them in a given ratio.

Example,
In the figure, P is the points that devides the line AB in the ratio 1:2.
As we move from A to B,

the x- coordinate increases by $9-3=6$.
the y- coordinate increases by $7-4=3$.



The changes in coordinates as we move from A to P is

$\frac{1}{3}$ of these.

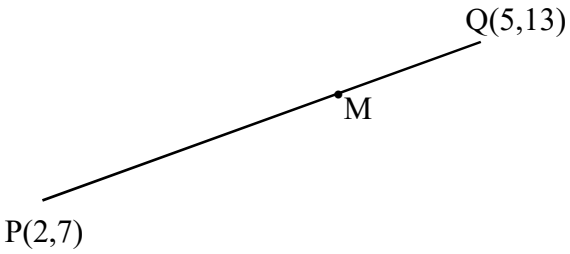
the x- coordinate increases by $6\times\frac{1}{3}=2$.

the y- coordinate increases by $3\times\frac{1}{3}=1$.

Then the coordinates of P,
x- coordinate = 3+3 = 6
y- coordinate = 4 +1 = 5

Coordinates of P are (6,5).

- In the figure, the point M devides the line PQ in different ratios.
  Complete the table.



PM:MQ	As we move from P to Q	As we move from P to M	x-coordinate of P	y-coordinate of P	coordinates of P
1:2	x-coordinate.... y-coordinate....	x-coordinate.... y-coordinate....			
1.2					
2:3					
3:5					

Line Maths

For any two points of a line, not parallel to either axis, the change in the y- coordinate is the change in the x- coordinate multiplied by a fixed number.

Example,
As we move from (4,5) to (7,11),
the change in the x- coordinate $= 7 - 4 = 3$
the change in the y- coordinate $= 11 - 5 = 6$

So,
the change in the y- coordinate is twice the the change in the x- coordinate.
Then the coordinates of two other points on this line,

(7+3, 11 +6) = (10,17)
(10+3, 17+6) = (13,23).

- Find the coordinates of two other points on the line joining each of the following pair of points.
  - (i) (2,0), (3,2)
  - (ii) (3,1), (5,4)
  - (iii) $(-2,3)$, $(1,5)$
  - (iv) $(-1,-2)$, (3,3)
  - (v) (0,3), (2,7)

Slope of a Line:

The slope of the line joining (x_1, y_1) and (x_2, y_2) $= \dfrac{y_2 - y_1}{x_2 - x_1}$

The slope of the line joining (3,5) and (6,7) $= \dfrac{7-5}{6-3} = \dfrac{2}{3}$

- Calculate the slope of the line joining each pair of points below.
  - (*i*) (0,2), (6,4)
  - (*ii*) (3,1), $(-2,6)$
  - (*iii*) (4,7), (6,10)
  - (*iv*) $(5,-3)$, $(4,-2)$
  - (*v*) (4,6), (6,3)
- Check whether the points (1,4), (5,2) and $(-3,6)$ are on the same line ?
- The slope of a line through (4,3) is $\dfrac{1}{2}$. Find the coordinates of two other points on this line.

- The slope of a line passing through (2,5) is $\frac{1}{3}$. Check whether the points given below are on the line or not.
  i) (5,6)
  ii) (3,8)
  iii) (9,-7)
  iv) (17,10)

Equation of a Line:

The relation between the coordinates of any point on the line can be written as an equation.

Example,

The slope of the line joining (3,5) and (6,7) is $\frac{7-5}{6-3}=\frac{2}{3}$

If a point of this line is taken as (x, y),

$\frac{y-5}{x-3}=\frac{2}{3}$

$3(y-5) = 2(x-3)$

$3y-15 = 2x-6$

$2x-3y + 15-6 = 0$

$2x-3y + 9 = 0$

Then the equation of the line is $2x-3y + 9 = 0$.

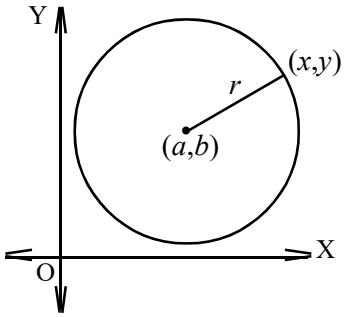
- Write the equation of the line joining each of the following pair of points.
  i) (3,0) ,(5,1)
  ii) (0,2) , (1,3)
  iii) (-1,2) , (6,4)
  iv) (5,3), (8,5)
  v) (3,5),(7,4)

- i) What is the slope of the line joining the points (2,4) and (5,10).
  ii) Write the equation of the line.

- i) Find the equation of the line joining (1,1) and (2,3).
  ii) Prove that if (x, y) is a point of this line, then so is $(x+1, y+2)$.

Equation of a Circle:

- The equation of the circle with the centre at the origin and radius r is $x^2+y^2=r^2$.
- The equation of the circle with the centre (a,b) and radius r is $(x-a)^2+(y-b)^2=r^2$.



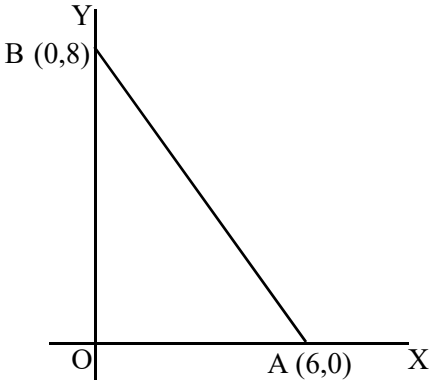
If (x,y) is a point on a circle with centre (2,3) and radius 5,
then the distance from the centre to the point (x,y) is the radius.
The square of the distance between the points (2,3) and (x,y) is,

$$(x-2)^2+(y-3)^2=5^2$$

$$(x-2)^2+(y-3)^2=25.$$

This is the equation of the circle with centre (2,3) and radius 5.

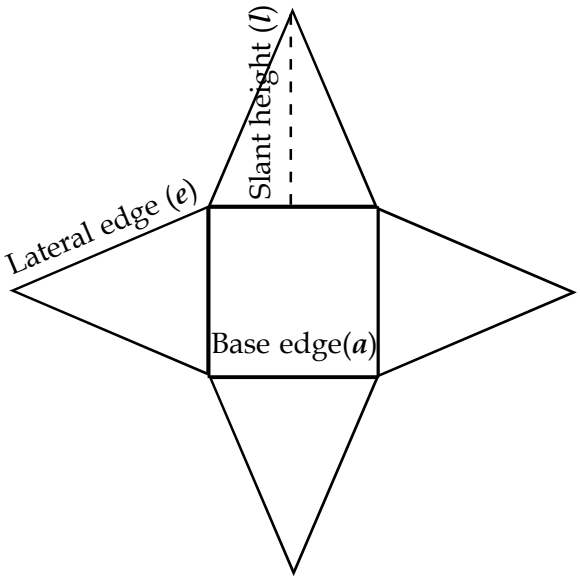
- Write the equation of the circle with centre at the origin and radius 3.
- i) Write the equation of the circle with centre at the origin and radius 4. Write
  ii) Write two other points on this circle.
- Find the equation of the circle with centre (3,4) and radius 8.
- Find the equation of the circle with centre $(-2,4)$ and radius 5.
- Find the equation of the circle with centre $(-1,-3)$ and radius 7.
- A circle is drawn with the line joining (1,2) and (7,10) as diameter.
  i) What are the copordinates of the centre of this circle ?
  ii) Write the equation of the circle.
- The equation of a circle is $(x-1)^2+(y-2)^2=16$.
  Find the coordinates of its centre and radius.
- The equation of a circle is $x^2+y^2-4x-6y-12=0$.
  Find the coordinates of its centre and radius.
- In the figure, the coordinates of the points A and B are (6,0) and (0,8). If AB is the diameter of the circle,
  i) Find the coordinates of the centre.
  ii) What is its radius ?
  iii) Write the equation of the circle.



12. SOLIDS

PYRAMID

A square pyramid is pyramid with a square shaped base.
The diagram given shows the unfolded shape of the square pyramid.



A square pyramid has 5 faces.
A square in the middle and 4 triangles around it, all four of them are isosceles triangles and they are equal.

A square pyramid has 8 edges.
4 base edges and
4 lateral edges.

If a square pyramid has base edge 'a', lateral edge'e', slant height 'l' and height 'h',

$$e^2 = l^2 + \left(\frac{a}{2}\right)^2$$

$$l^2 = h^2 + \left(\frac{a}{2}\right)^2$$

Surface Area:

If a square pyramid has base edge 'a'and slant height 'l'
Base area $= a^2$
Lateral surface area $= 2al$
Total surface area = base area + lateral surface area = $a^2 + 2al$

Volume of a Pyramid:

The volume of a square pyramid is equal to a third of the product of the base area and the height.

Volume of a square pyramid $= \frac{1}{3} \times$ base area $\times$ height

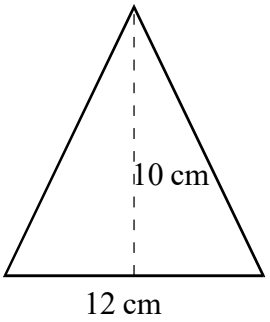
$= \frac{1}{3}a^2h$

More Questions:

1) The base edge of a square pyramid is 12 centimetres and lateral edge 10 centiimetres. What is the slant height ?
2) The slant height of a square pyramid is 13 centimetres and base edge 10 centiimetres. What is the height of the pyramid ?
3) The diamentions of a square pyramids are given below in centimetres. Fill the table.

Base edge (a)	Slant height (l)	Lateral edge (e)	Height (h)
6	...	5	...
...	10	...	8
10	...	...	12
...	12	15	...

4) One lateral surface of a square pyramid is in the figure.
   (a) What is the base edge of the square pyramid ?
   What is the slant height ?
   (b) What is the base area of the square pyramid ?
   (c) What is the lateral surface area ?
   (d) What is the total surface area ?



5) The base edge of a square pyramid is 8 centimetres and lateral edge 17 centiimetres.
   (a) What is the slant height ?
   (b) What is its base area ?
   (c) What is its lateral surface area ?
   (d) What is the total surface area ?

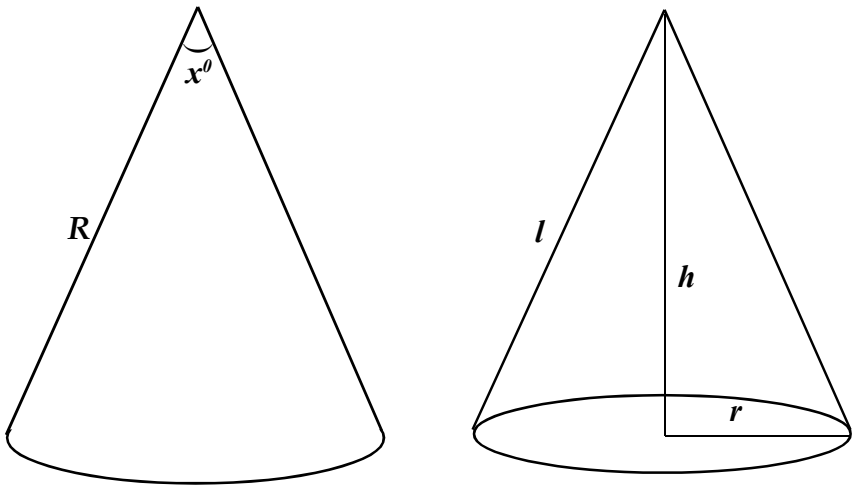
6) The perimeter of the base of asquare pyramid is 40 centimetres and the height 12 centimetres.
   (a) What is the length of the base edge ?
   (b) What is the slant height ?
   (c) What is the total surface area ?

7) The base edge of a square pyramid is 5 centimetres and the height is 12 centimetres.
   (a) What is its base area ?
   (b) Find the volume of the pyramid.

8) Complete the table.

Base edge cm.	Height cm.	slant height cm.	base area cm^2.	Volume cm^3.
10	...	13	...	...
...	8	...	144	...
...	9	15	...	...
40	21	...	...	...

Cone:



Cone made by rolling up a sector of a circle,

- The radius of the sector becomes the slant height of the cone ($R=l$).
- The arc length of the sector becomes the circumference of the base of the cone.

The central angle of the sector x^o, base radius of the cone 'r' and slant height 'l',

$$\frac{x^0}{360} = \frac{r}{l}$$

If height of the cone 'h', base radius 'r' and slant height 'l',

$$l^2 = h^2 + r^2$$

Surface area:

Curved surface area = πrl

Base area = πr^2

Surface area = base area + curved surface area

= $\pi r^2 + \pi rl$

Volume of a cone:

The volume of a cone is equal to a third of the product of the base area and height.

Volume of a cone $= \frac{1}{3} \times$ base area $\times$ height

If the base radius 'r' and height 'h',

Volume $= \frac{1}{3} \times \pi r^2 \times h$

More Questions:

1) What are the radius of the base and slant height of a cone made by rolling up a sector of central angle 60^0 cut out from a circle of radius 12 centimetres ?

2) What are the radius of the base and slant height of a cone made by rolling up a sector of central angle 120^0 cut out from a circle of radius 18 centimetres ?

3) The diameter of the base of a cone is 12 centimetres and height 8 centimetres.
   (a) What is its slant height ?
   (b) What is its curved surface area ?
   (c) What is its base area ?
   (d) What is its total surface area ?

4) From a circle of radius 12 centimetres a sector of central angle 120^0 is cut out and make a cone.
   (a) What is the slant height of the cone ? What is its radius ?
   (b) What is the curved surface area of the cone ?
   (c) What is its total surface area ?

5) From a circular metal sheet of radius 15 centimetres, a sector of central angle 72^0 is cut out and rolled up into a cone.
   (a) What is the slant height of the cone ?
   (b) What is the curved surface area of the cone ?

6) The base radius of a cone is 6 centimetres and its height is 8 centimetres.
   Find its volume.

7) Complete the table given below.

Base radius cm.	Height cm.	slant height cm.	base area cm^2.	Volume cm^3.
5	12	...	...	...
...	8	10	...	...
9	...	15	...	...
15	20	...	...	...

Sphere:

The surface area of sphere with radius 'r' $= 4\pi r^2$

$$\text{Volume} = \frac{4}{3}\pi r^3$$

The surface area of hemisphere with radius 'r' $= 3\pi r^2$

$$\text{Volume} = \frac{2}{3}\pi r^3$$

More Questions:

1) The diameter of a sphere is 20 centimetres. Find its total surface area and its volume.
2) The total surface area of a solid sphere is 120π square centimetres. It is cut into two halfs. Find the surface area of each hemisphere.
3) A metal sphere of radius 9 centimetres, melted and recast into small spheres of radius 3 centimetres. How many spheres can be made ?
4) Ratio of the diameter of two hemisphere is 2:5.
   (i) Find the ratio of their radius.
   (ii) Find the ratio of their surface area.
   (iii) If the surface area of the second hemisphere is 50 square centimetres. Calculate the surface area of the first hemisphere.
5) A largest sphere is carved out from a cube. If the volume of the sphere is 288π cubic centimetres, find the volume of the cube.

13. STATISTICS

MEDIAN:

If the given numbers are arranged in increasing order, the number that comes in the middle is the Median.

If the number of the given data is an even number, there will be two numbers in the middle. In that case the median is found by taking half of the sum of those two numbers.

Activities:

- Find the median of the numbers
  12, 5, 18, 14, 22, 6, 10

  Write the given numbers in increasing order.

  5, 6, 10, (12) 14, 18, 22

  Number comes in the middle : 12

  So, Median : 12

- The scores obtained by 10 student in a Mathematics Exam are given below.
  32, 41, 20, 15, 42, 50, 36, 40, 45, 38

  Write the numbers in increasing order,

  15, 20, 32, 36, [38, 40,] 41, 42, 45, 50

  Nnumbers coming in the middle : 38, 40

  $$\text{Median} = \frac{38+40}{2} = \frac{78}{2} = 39$$

More Activities:

1) Find the median of the following data.
   (i) 5, 10, 18, 9, 16, 25, 22
   (ii) 42, 38, 36, 50, 40, 32, 43, 29
   (iii) 102, 105, 120, 106, 125, 132, 121, 130, 124, 112
   (iv) 18, 21, 17, 24, 20
2) The weights of 7 children are given.
   32, 40, 38, 42, 30, 34, 38
   Find the median weight.
3) Find the median of first 9 even numbers.
4) Scores of 10 students obtained in an exam are given below.
   30, 28, 25, 32, 20, 36, 24, 33, 27, 38
   Find the median score.

Frequency and Median:

Activities:

Finding median using a frequency table .

Weight (in Kg.)	Number of persons
30	3
40	6
50	8
60	12
70	9
80	3
Total	41

Calculate the median weight from the given table.

The median is the value that comes in the middle. So, out of the total 41 persons, 20 persons weight should be less than the median and 20 persons weight should be more than the median.

If we arrange the 41 people in order of their weights, the weight of the 21st person is the median.

To calculate this, we write the cumulative frequency up to each range, using the given table.

Weight (in Kg.)	Number of persons
upto 30	3 = 3
upto 40	3+6 = 9
upto 50	3+6+8 = 17
upto 60	3+6+8+12 = 29
upto 70	3+6+8+12+9 = 38
upto 80	3+6+8+12+9+3 = 41

From the table, we can see that they weights of persons from 17 to 29 are 60.
So, the weight of 21st person is also 60.

Median weight = 60.

More Activities:

1) The table below shows the monthly income of 45 households. Find the median income.

Monthly Income (in Rs.)	Number of Households
5000	4
5500	6
6000	8
6500	10
7000	9
7500	7
8000	1

2) The table below shows the daily wages of some workers in a company.

Daily wages (in Rs.)	Number of workers
650	5
750	8
850	9
950	11
1050	7
1150	6

Find the median income.

Classes and Median:

The table shows members in a Kudumbasree unit sorted according to their ages.

Age	Number of Members
20 - 30	5
30 - 40	10
40 - 50	5
50 - 60	8
60 - 70	4
70 - 80	3
Total	35

Find the median age.

If we arrange the members from the youngest to oldest by age, we need to find the age of 18th person who falls in the middle.

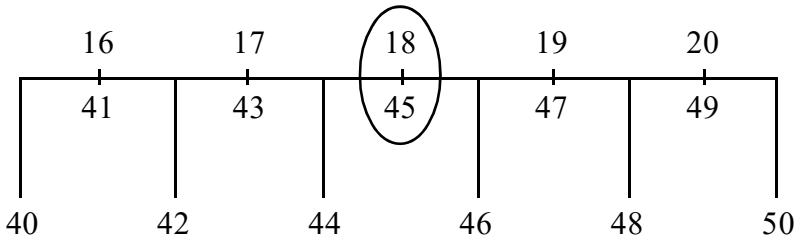
To find which group the 18th person belongs to, let's prepare a table adding up the members in each class.

Age	Number of members
Below 30	5 = 5
Below 40	5+10=15
Below 50	5+10+5 = 20
Below 60	5+10+5+8 = 28
Below 70	5+10+5+8+4 = 32
Below 80	5+10+5+8+4+3 = 35

From the table, it is clear that the ages of the five members from 16th to 20th fall between 40 and 50.

So, the age of the 18th member is also between 40 and 50.

To calculate this more accurately, divide the 10 units between 40 and 50 into 5 equal classes.



From this, we can see that the age of the 18th person is 45.

Median age = 45

Alternative Method:

This can also be calculated in another way.

Assume, the ages of members from 16th to 20th are terms in an arithmetic sequence.

Age of the 16th member (x_{16}) = 41

Common difference $= \frac{10}{2} = 5$

Age of the 18th member (x_{18}) $= 41 + 2 \times 2$

$= 41+4 = 45$

More Activities:

1) The table shows the workers in factory sorted according to their ages.

Age	Number of workers
20 - 30	9
30 - 40	10
40 - 50	8
50 - 60	5
60 - 70	1

(i) If the workers are arranged in increasing order of their ages, the age of the worker at what position is taken as the median ?

(ii) Find the median age.

2) The table below shows the workers in a factory sorted according to their daily wages.

Daily wage (in Rs.)	Number of workers
800 - 900	5
900 - 1000	7
1000 - 1100	6
1100 - 1200	10
1200 - 1300	15
1300 - 1400	2

(i) If the workers are arranged in increasing order of daily wages, what is assumed to be the daily wage of the worker at the 19th position.

(ii) Find the median daily wage.

3) The table shows some households sorted according to their usage of electricity.

Usage of Electricity	Number of households
80 - 100	8
100 - 120	12
120 - 140	10
140 - 160	9
160 - 180	6

(i) If the households are arranged in increasing order of their electricity usage, which household is taken as the median ?

(i) Which is the median class ?

(i) Find the median usage.